Enterprise Synthesis Team

PHARMACY MANAGEMENT SYSTEM

SRS

SOFTWARE REQUIREMENTS SPECIFICATION

SRS • Initial Release • Four-Month Academic Software Project

DOCUMENT	Software Requirements Specification (SRS)
PROJECT	Pharmacy Management System (PMS)
PROJECT LEADER	Bahram Jafaree
DURATION	4 Months
STATUS	BASELINE FOR DEVELOPMENT

Prepared for analysis, design, implementation, verification and acceptance

00 Document Control

Governance, ownership and requirements baseline

Field	Value
Document Title	Software Requirements Specification (SRS)
System	Pharmacy Management System (PMS)
Version	1.0
Project Duration	4 Months
Project Type	Software Development Project
Team Leader	Bahram Jafaree
Document Status	Initial Baseline

Development Team

Name	Role
Bahram Jafaree	Team Leader / Project Manager
Said Yousaf Hashimi	System Designer & Analyst / Backend Developer
Ali Seena	Frontend / UI Developer
Alireza Alizada	Database Developer
Zaki Ahmadi	QA & Documentation

SRS PURPOSE

This document is the agreed technical baseline for what the initial PMS must do, how it should behave, the constraints it must respect, and the conditions under which it will be accepted.

01 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Pharmacy Management System (PMS).

The PMS will provide a centralized software solution for managing the daily operations of a pharmacy, including medicine management, inventory, sales, purchases, suppliers, users and reporting.

This document will serve as a reference for the project team during analysis, design, development, testing and implementation.

1.2 Scope of the System

The Pharmacy Management System will be designed to automate and simplify pharmacy operations. Authorized users will be able to:

- Manage medicines.
- Manage medicine categories.
- Monitor inventory.
- Track medicine expiration dates.
- Record medicine purchases.
- Record medicine sales.
- Manage suppliers.
- Manage system users.
- Search and filter information.
- Generate reports.
- View important information through a dashboard.

The system will reduce dependence on manual records and help improve the accuracy, efficiency and organization of pharmacy operations.

02 Overall Description

2.1 Product Perspective

The PMS will be a centralized management application consisting of several interconnected modules. The modules will use a common database so that information can be shared consistently throughout the system.

No.	Module
1	Authentication and User Management
2	Medicine Management
3	Category Management
4	Inventory Management
5	Sales Management
6	Purchase Management

7	Supplier Management
8	Dashboard
9	Reporting
10	System Administration

03 Product Functions

High-level capability map

3.1 Authentication

Users will be required to log in before accessing protected parts of the system.

3.2 Medicine Management

Authorized users will be able to:

- Add medicines.
- Update medicines.
- Delete/deactivate medicines.
- Search medicines.
- View medicine details.
- Assign medicines to categories.
- Store batch information.
- Store expiration dates.
- Store buying and selling prices.
- Store available quantities.

3.3 Inventory Management

The system will:

- Track medicine quantities.
- Increase stock after purchases.
- Decrease stock after sales.
- Identify low-stock medicines.
- Identify expired medicines.
- Identify medicines approaching expiration.
- Maintain inventory records.

3.4 Sales Management

Users will be able to:

- Create sales transactions.
- Search for medicines.
- Select quantities.
- Calculate subtotals.
- Calculate total sales.

- Record sales.
- Generate a receipt/invoice.
- View previous sales.

3.5 Purchase Management

Users will be able to:

- Create purchase records.
- Select suppliers.
- Add medicines to purchases.
- Record purchase quantities.
- Record purchase prices.
- Update inventory automatically.
- View purchase history.

3.6 Supplier Management

Authorized users will be able to:

- Add suppliers.
- Edit supplier information.
- Deactivate suppliers.
- Search suppliers.
- View supplier information.
- View supplier purchase history.

3.7 Reporting

The system will generate:

- Daily sales reports.
- Monthly sales reports.
- Purchase reports.
- Inventory reports.
- Low-stock reports.
- Expired medicine reports.
- Near-expiration reports.
- Medicine transaction history.

04 User Classes and Characteristics

4.1 Administrator

The administrator will have the highest level of access.

- Manage users.
- Manage roles and permissions.
- Manage medicines.
- Manage inventory.

- View reports.
- Manage suppliers.
- Monitor system activities.

4.2 Pharmacist / Manager

- Manage medicines.
- View inventory.
- Process sales.
- Record purchases.
- Manage suppliers.
- View reports.

4.3 Sales / Cashier Staff

- Search medicines.
- Process sales.
- Generate receipts.
- View relevant sales information.

ACCESS LIMITATION

The cashier will not have access to sensitive administrative functions unless permission is granted.

05 Functional Requirements

What the system must do

FR-01

User Login

The system shall allow registered users to log in using valid credentials.

- The user shall enter a username/email and password.
- The system shall validate the credentials.
- The system shall reject invalid credentials.
- The system shall redirect authenticated users to the appropriate dashboard.
- The system shall provide a logout function.

FR-02

User Management

The administrator shall be able to:

- Create user accounts.
- Edit user accounts.
- Deactivate user accounts.
- Assign roles.
- Reset/change passwords where permitted.
- View registered users.

FR-03

Role-Based Access Control

The system shall restrict functionality according to user roles.

Function	Admin	Manager / Pharmacist	Cashier
Manage Users	Yes	—	—
Manage Medicines	Yes	Yes	View
Manage Inventory	Yes	Yes	—
Process Sales	Yes	Yes	Yes
Manage Purchases	Yes	Yes	—
Manage Suppliers	Yes	Yes	—
View Reports	Yes	Yes	Limited
System Settings	Yes	—	—

06 Medicine Management Requirements

FR-04

Add Medicine

The system shall allow authorized users to add a new medicine.

- Medicine ID.
- Medicine name.
- Generic name.
- Category.
- Manufacturer.
- Batch number.
- Expiration date.
- Purchase price.
- Selling price.
- Quantity.
- Minimum stock level.
- Description.

FR-05

Update Medicine

The system shall allow authorized users to update existing medicine information.

FR-06

Search Medicine

The system shall allow users to search medicines by:

- Medicine name.
- Generic name.
- Category.
- Manufacturer.
- Batch number.

FR-07

Delete / Deactivate Medicine

The system shall allow authorized users to deactivate medicines that are no longer available.

- The system should avoid permanently deleting medicines associated with historical sales or purchase records.

07 Category Management

FR-08

Manage Categories

Authorized users shall be able to:

- Add medicine categories.
- Edit categories.
- Deactivate categories.
- View categories.

EXAMPLES

Antibiotics • Painkillers • Vitamins • Syrups • Tablets • Capsules • Injections

08 Inventory Requirements

FR-09

View Stock

The system shall display the current quantity of medicines.

FR-10

Automatic Stock Update

The system shall automatically:

- Increase stock when a purchase is recorded.
- Decrease stock when a sale is completed.

FR-11

Low Stock Alert

The system shall identify medicines whose quantity has reached or fallen below the defined minimum stock level.

FR-12

Expiration Monitoring

The system shall identify:

- Expired medicines.
- Medicines approaching expiration.

FR-13

Inventory History

The system shall maintain a history of inventory changes, including:

- Date.

- Medicine.
- Quantity changed.
- Change type.
- Related transaction.
- User responsible.

09 Sales Requirements

FR-14

Create Sale

Authorized users shall be able to create a sales transaction.

- Search for a medicine.
- Select the medicine.
- Enter quantity.
- Calculate price.
- Add additional medicines.
- Calculate the total.
- Confirm the sale.

FR-15

Validate Stock

The system shall verify that sufficient stock exists before completing a sale.

- The system shall not allow a user to sell more units than the available quantity.

FR-16

Sales Receipt

After completing a sale, the system shall generate a sales receipt/invoice containing relevant information such as:

- Transaction ID.
- Date/time.
- Medicine name.
- Quantity.
- Unit price.
- Total price.

FR-17

Sales History

Authorized users shall be able to view previous sales transactions.

- Search/filter by date.
- Search/filter by transaction ID.
- Search/filter by medicine.
- Search/filter by user.

10 Purchase Requirements

FR-18

Record Purchase

Authorized users shall be able to record medicine purchases.

- Purchase ID.
- Supplier.
- Medicine.
- Quantity.
- Purchase price.
- Date.
- Batch number.
- Expiration date.

FR-19

Update Inventory After Purchase

When a purchase is successfully recorded, the system shall automatically update the corresponding medicine stock.

FR-20

Purchase History

The system shall maintain purchase history and allow authorized users to search previous purchases.

11 Supplier Requirements

FR-21

Add Supplier

The system shall allow authorized users to add suppliers.

- Supplier ID.
- Supplier name.
- Company name.
- Phone number.
- Email.
- Address.
- Description.

FR-22

Update Supplier

Authorized users shall be able to modify supplier information.

FR-23

Supplier History

The system shall allow authorized users to view purchase transactions associated with a supplier.

12 Dashboard Requirements

FR-24

Dashboard

The system shall provide a dashboard containing important pharmacy information.

- Total medicines.
- Total categories.
- Total suppliers.
- Current inventory.
- Low-stock medicines.
- Expired medicines.
- Today’s sales.
- Monthly sales.
- Recent transactions.

VISUALIZATION STANDARD

The dashboard should present information using cards, tables, charts or other appropriate visual elements.

13 Reporting Requirements

FR-25

Sales Reports

The system shall generate sales reports based on selected dates.

FR-26

Purchase Reports

The system shall generate purchase reports based on selected dates.

FR-27

Inventory Report

The system shall provide a report showing current inventory.

FR-28

Expiration Report

The system shall provide information about expired and near-expiration medicines.

FR-29

Low Stock Report

The system shall generate a list of medicines requiring restocking.

14 Search and Filtering Requirements

FR-30

Search

The system shall provide search functionality for major data entities.

- Medicines.
- Suppliers.
- Sales.
- Purchases.

- Users.
- Categories.

15 Non-Functional Requirements

How well the system should work

NFR-01

Performance

The system should respond to normal user operations quickly.

- For standard operations, the target response time should generally be within 2–3 seconds, depending on deployment environment and network conditions.

NFR-02

Security

The system shall:

- Require authentication for protected functions.
- Use secure password storage.
- Implement role-based authorization.
- Validate user input.
- Protect sensitive information.
- Prevent unauthorized access.

NFR-03

Usability

The system should:

- Have a simple interface.
- Use clear labels.
- Provide understandable error messages.
- Have consistent navigation.
- Minimize unnecessary steps.
- Be easy for pharmacy employees to learn.

NFR-04

Reliability

The system should operate reliably during normal pharmacy operations.

- The system should prevent data corruption and maintain consistent transaction records.

NFR-05

Availability

The system should be available during pharmacy working hours, subject to the reliability of the hosting/server environment.

NFR-06

Maintainability

The software should have:

- Organized source code.

- Modular architecture.
- Meaningful variable/function names.
- Documentation.
- Version control.
- Clearly separated components.

NFR-07

Scalability

The system should be designed so that additional functionality can be added in future versions without completely redesigning the system.

- Multi-branch support.
- Barcode scanning.
- Online ordering.
- Mobile application.
- Advanced analytics.
- Payment integration.

NFR-08

Backup and Recovery

The database should be backed up regularly.

- The system should provide an appropriate recovery mechanism in case of data loss or system failure.

16 External Interface Requirements

16.1 User Interface

The PMS should provide interfaces for:

- Login.
- Dashboard.
- Medicine management.
- Inventory.
- Sales.
- Purchases.
- Suppliers.
- Users.
- Reports.
- Settings.

The interface should be responsive and easy to navigate.

16.2 Hardware Interface

The initial version will operate using standard computers/laptops. Optional future hardware integrations may include:

- Barcode scanner.
- Receipt printer.
- Label printer.

INITIAL RELEASE BOUNDARY
These hardware integrations are not mandatory for the initial version.

16.3 Software Interface

The system will interact with:

- Database management system.
- Operating system.
- Web browser, if implemented as a web application.
- Backend server/application framework.

17 Database Requirements

The system will maintain a centralized database.

17.1 Main Tables

Entity / Table	Purpose
Users	Core PMS data entity
Roles	Core PMS data entity
Medicines	Core PMS data entity
Categories	Core PMS data entity
Suppliers	Core PMS data entity
Purchases	Core PMS data entity
Purchase Items	Core PMS data entity
Sales	Core PMS data entity
Sale Items	Core PMS data entity
Inventory	Core PMS data entity
Inventory Transactions	Core PMS data entity

17.2 Basic Relationships

Relationship	Cardinality / Meaning
Supplier → Purchases	One supplier can have multiple purchases.
Purchase → Purchase Items	One purchase can contain multiple medicines.
Medicine → Purchase Items	A medicine can appear in multiple purchase records.
Sale → Sale Items	One sale can contain multiple medicines.
Medicine → Sale Items	A medicine can appear in multiple sales.
Medicine → Category	A category can contain multiple medicines.

18 Use Case Overview

Administrator

- Login.
- Manage users.
- Manage medicines.
- Manage categories.
- Manage inventory.
- Manage suppliers.
- Manage purchases.
- Manage sales.
- View reports.
- Manage system settings.

Pharmacist / Manager

- Login.
- Manage medicines.
- Manage inventory.
- Manage suppliers.
- Record purchases.
- Process sales.
- View reports.

Cashier

- Login.
- Search medicines.
- Process sales.
- Generate receipts.
- View permitted sales information.

19 Use Case Table

Use Case	Actor	Description
Login	All Users	Authenticate into the system
Manage Users	Admin	Create, update and deactivate users
Manage Medicines	Admin / Manager	Add, edit, search and deactivate medicines
Manage Categories	Admin / Manager	Manage medicine categories
Manage Inventory	Admin / Manager	Monitor and update stock
Record Purchase	Admin / Manager	Record medicine purchases
Manage Suppliers	Admin / Manager	Manage supplier information
Process Sale	Admin / Manager / Cashier	Complete customer sales
Generate Receipt	Admin / Manager / Cashier	Generate sale receipt
View Reports	Admin / Manager	Generate and view reports
Monitor Expiration	Admin / Manager	Identify expired medicines
Monitor Low Stock	Admin / Manager	Identify medicines requiring restocking

20 Business Rules

1. Only authenticated users can access protected functions.
2. Users can only perform operations permitted by their roles.
3. A medicine must have a unique identifier.
4. Medicine quantity cannot normally be negative.

5. A sale cannot be completed when sufficient stock is unavailable.
6. Completed sales must update inventory.
7. Completed purchases must update inventory.
8. Expired medicines should be identified by the system.
9. Historical transactions should not be unnecessarily deleted.
10. Important database records should be backed up regularly.
11. Required medicine information must be validated before saving.
12. Every transaction should have a unique transaction identifier.

21 Data Validation Requirements

The system shall validate data before saving it.

- Required fields cannot be empty.
- Phone numbers should follow an acceptable format.
- Email addresses should follow a valid format when provided.
- Prices cannot be negative.
- Quantities cannot be negative.
- Expiration dates must be valid dates.
- Usernames/emails should be unique where applicable.
- Medicine IDs should be unique.
- Transaction IDs should be unique.

22 Error Handling

The system shall provide understandable error messages and should avoid exposing technical/database errors directly to normal users.

Scenario	User-Facing Message
Invalid login	“Invalid username or password.”
Insufficient stock	“Insufficient stock. Available quantity: 5.”
Missing information	“Please enter the medicine name.”
Duplicate record	“A medicine with this ID already exists.”

23 Security Requirements

Control Area	Requirement

Control Area	Requirement
Authentication	Users must authenticate before accessing protected resources.
Authorization	Each user shall have permissions according to their role.
Password Security	Passwords should not be stored as plain text.
Input Security	User inputs should be validated and sanitized to reduce security vulnerabilities.
Database Security	Only authorized application components should access the database.
Session Security	User sessions should be appropriately managed and terminated when users log out.

24 Assumptions

1. Users will have basic computer knowledge.
2. Pharmacy staff will provide accurate information.
3. Required hardware and software will be available.
4. The database server will be available during system operation.
5. The project will be completed within four months.
6. The initial system will focus on essential pharmacy-management functions.
7. Major changes to requirements will require review and approval.
8. Users will participate in testing and provide feedback.

25 Constraints

- Four-month development period.
- Limited team size.
- Limited project resources.
- Limited development time.
- Possible changes in requirements.
- Limited access to real-world pharmacy data.
- Advanced integrations are outside the initial scope.

26 Future Enhancements

Version 2

- Barcode scanning.

- Automatic receipt printing.
- Multi-branch management.
- Customer management.
- Advanced inventory analytics.

Version 3

- Mobile application.
- Online medicine ordering.
- Online delivery management.
- Payment gateway integration.
- Supplier portal.

Advanced Version

- AI-powered sales/inventory forecasting.
- Demand prediction.
- Automated restocking recommendations.
- Advanced business analytics.
- Cloud-based multi-branch synchronization.

27 Acceptance Criteria

- Users can successfully authenticate.
- Roles and permissions work correctly.
- Medicines can be managed.
- Categories can be managed.
- Inventory is accurately maintained.
- Purchases can be recorded.
- Sales can be processed.
- Stock updates automatically after transactions.
- Suppliers can be managed.
- Expired and low-stock medicines can be identified.
- Required reports can be generated.
- Major security requirements are implemented.
- Critical bugs have been resolved.
- The system passes the agreed testing process.
- Documentation is completed.
- The system can be demonstrated successfully.

ACCEPTANCE PRINCIPLE

Acceptance is based on successful implementation, verification and demonstration of the approved requirements—not simply completion of coding.

28 Requirements Traceability

A simple requirements traceability structure will be used to ensure that every requirement is designed, implemented and tested.

Requirement	Design	Development	Testing
User Login	Authentication Design	Login Module	Login Test
Medicine Management	Medicine Module	Medicine Module	Medicine Tests
Inventory	Inventory Design	Inventory Module	Inventory Tests
Sales	Sales Design	Sales Module	Sales Tests
Purchases	Purchase Design	Purchase Module	Purchase Tests
Suppliers	Supplier Design	Supplier Module	Supplier Tests
Reports	Reporting Design	Reporting Module	Report Tests
User Roles	Security Design	Authorization	Permission Tests

29 SRS Approval

APPROVAL STATEMENT

This SRS represents the agreed requirements baseline for the initial Pharmacy Management System. Approved changes should be managed through the project change-control process.

Name	Role	Signature
Bahram Jafaree	Team Leader	_____
Said Yousaf Hashimi	System Analyst / Backend Developer	_____
Ali Seena	Frontend / UI Developer	_____
Alireza Alizada	Database Developer	_____
Zaki Ahmadi	QA & Documentation	_____
Project Supervisor	Supervisor	_____

Date: _____

END OF SOFTWARE REQUIREMENTS SPECIFICATION

Pharmacy Management System • SRS v1.0 • Proposed